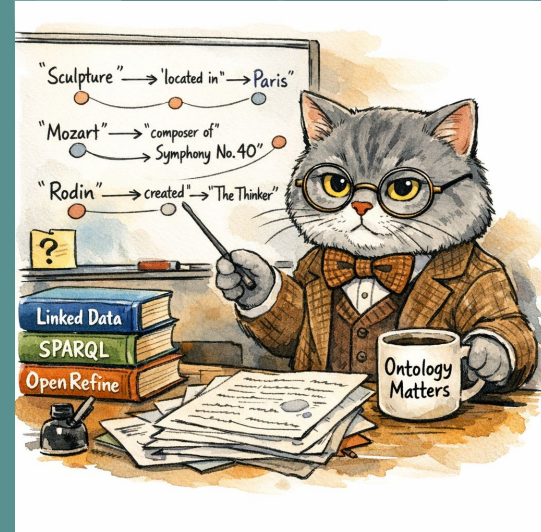
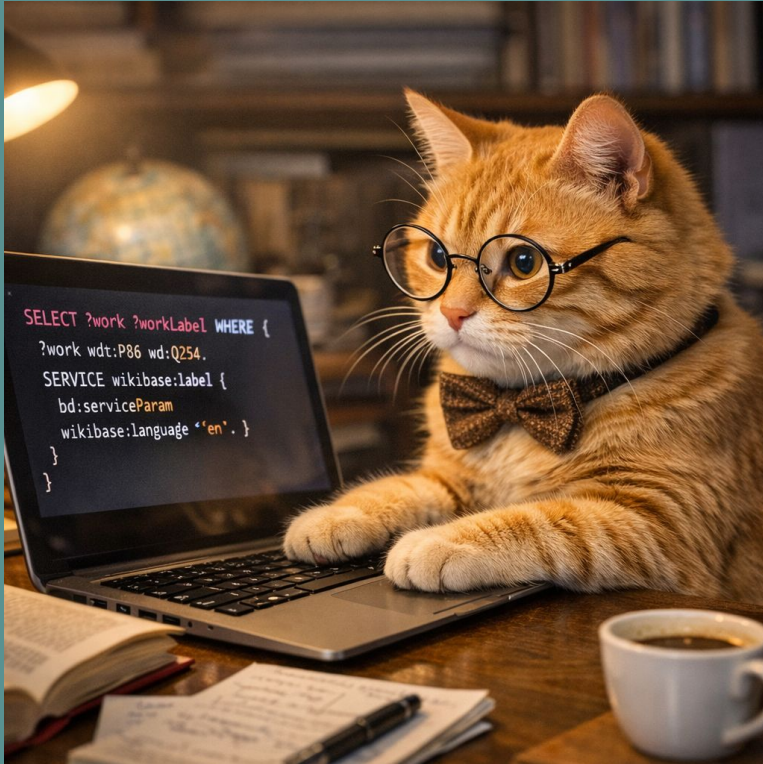


Examination – Practical Part



Funny cat images provided by Microsoft Copilot and the web ♡



Exercise 1 – Wikidata



Displaying Results

When viewing the Wikidata page for Wolfgang Amadeus Mozart, several key statements appear in the item's structured data. These include his entity type ("instance of: human"), his place of birth (Salzburg), his parents (Leopold Mozart and Anna Maria Mozart), and notable works such as *The Magic Flute*, *Don Giovanni*, and *Eine kleine Nachtmusik*. Each value is linked to a unique Q-number, and each property is identified by a P-number.

Analysis of Results

These results show how Wikidata uses an RDF-like structure:

- **Q-numbers** uniquely identify items (e.g., Q5 = "human", Q34713 = "Salzburg").
- **P-numbers** identify properties (e.g., P31 = "instance of", P19 = "place of birth").

This creates a shared, unambiguous vocabulary. Because every item and property has a stable identifier, data from different sources can be merged, compared, and queried without confusion. For example, if two datasets both use Q34713 for Salzburg, they can be combined reliably.

Question(s)	Wikidata Property	Q-Number
The entity type(s) of Mozart	instance of human P31	Q5
Mozart's place of birth	place of birth Salzburg P19	Q34713
Mozart's parents	father Leopold Mozart P22	Q156280
	mother Anna Maria Mozart P25	Q167981
At least five musical compositions	notable work The Magic Flute The Marriage of Figaro Don Giovanni Cosi fan tutte Die Entführung aus dem Serail Symphony No. 40 Eine kleine Nachtmusik Requiem P800	 Q5064 Q201873 Q192039 Q207410 Q210711 Q231390 Q12025 Q207875

Exercise 2 – Querying Wikidata using SPARQL



Queries Performed

- A query using **P86 (composer)** returned 944 musical works composed by Mozart.
- Combining **P31 (instance of: opera)** with **P86** returned no results, showing that not all operas are consistently typed as “opera” in Wikidata.
- Using **P737 (influenced by)** returned 13 people influenced by Mozart, including Beethoven and Chopin.
- Filtering by **P1191 (date of first performance)** for the year 1787 returned two works: *Don Giovanni* and *Symphony No. 38*.

Analysis of Results

- SPARQL queries RDF data by matching patterns of triples (subject–predicate–object).
- Wikidata uses **Q-numbers** for items and **P-numbers** for properties, creating a stable, unambiguous vocabulary.
- This structure supports Linked Open Data because datasets using the same identifiers can be merged and queried consistently.
- SPARQL’s query types (ASK, SELECT, CONSTRUCT, DESCRIBE) allow flexible filtering and retrieval, making it easy to explore large datasets such as Mozart’s works.
- Although data quality still depends on contributors, the use of shared identifiers enables reliable linking, reconciliation, and SPARQL querying.

Question	Query	Results	Properties Used
Find all musical works composed by Wolfgang Amadeus Mozart.	<pre> SELECT ?work ?workLabel WHERE { ?work wdt:P86 wd:Q254 . SERVICE wikibase:label { bd:serviceParam wikibase:language "[AUTO_LANGUAGE],en". } }</pre>	944 results were returned, displaying Mozart's works in multiple languages, each linked to its own Wikidata page via its unique Q-number.	P86 denotes the property "composer".
Retrieve all operas composed by Mozart.	<pre> SELECT ?opera ?operaLabel WHERE { ?opera wdt:P31 wd:Q1344 ; # instance of opera wdt:P86 wd:Q254 . # composer Mozart SERVICE wikibase:label { bd:serviceParam wikibase:language "[AUTO_LANGUAGE],en". } }</pre>	This returns no results.	P31 denotes the property "instance of", and P86 denotes "composer".
Find people influenced by Mozart.	<pre> SELECT ?person ?personLabel WHERE { ?person wdt:P737 wd:Q254 . SERVICE wikibase:label { bd:serviceParam wikibase:language "[AUTO_LANGUAGE],en". } }</pre>	13 results were returned, including Beethoven and Chopin.	P737 denotes the property "influenced by".

Find all pieces composed by Mozart first performed in 1787.	<pre> SELECT ?work ?workLabel ?date WHERE { ?work wdt:P86 wd:Q254 ; # composer = Mozart wdt:P1191 ?date . # date of first performance FILTER(?date >= "1787-01-01T00:00:00Z"^^xsd :dateTime && ?date < "1788-01-01T00:00:00Z"^^xsd :dateTime) SERVICE wikibase:label { bd:serviceParam wikibase:language "[AUTO_LANGUAGE],en". } } </pre>	This returns two results: <i>Don Giovanni</i>, performed on 29 October 1787, and <i>Symphony No. 38</i>, performed on 19 January 1787.	P86 denotes “composer”, and P1191 denotes “date of first performance”.
--	--	---	---



```
1 SELECT ?person ?personLabel WHERE {  
2   ?person wdt:P737 wd:Q254.  
3   SERVICE wikibase:label { bd:serviceParam wikibase:language "[AUTO_LANGUAGE],en". }  
4 }  
5
```

Table ?

person	personLabel
 wd:Q255	Ludwig van Beethoven
 wd:Q1268	Frédéric Chopin
 wd:Q7294	Johannes Brahms



Exercise 3 – Named Entity Recognition

Tool(s) Used

SpaCy NER demo, because my OpenRefine installation didn't support the NER extension. I copied sample rows into the demo to extract entities and analysed the outputs in OpenRefine.

Analysis

- **Entity types detected:** PERSON, ORG, GPE
- **Accuracy:** Strong for PERSON and GPE; weaker for ORG and ambiguous names
- **False positives:** Titles and locations occasionally misclassified
- **Usefulness:** Helps identify creators and places for linking to Wikidata
- **Limitations:** Context sensitivity — titles and compound names can confuse the model



Implications

NER highlights the need for domain-specific tuning in cultural-heritage datasets and supports interoperability when combined with reconciliation workflows. Even imperfect extraction adds semantic structure to unstructured text, enabling links between descriptions and structured knowledge graphs and improving discoverability across Linked Open Data.



Column	Values	Results
title	Fontaine aux roseaux d'argent fontaine du Fella <u>fontaine</u> du Pot-de-Fer	Fontaine aux roseaux d'argent fontaine du Fella PERSON fontaine du Pot-de-Fer PERSON
creator	Louis Petitot Jean-Pierre Dantan, Antoine Laurent Dantan, Édouard Renaud Pierre Granet	Louis Petitot PERSON Jean-Pierre Dantan PERSON, Antoine Laurent ORG, Dantan ORG, Édouard Renaud Pierre Granet PERSON
genre	fountain statue, fountain statue	fountain statue, fountain statue

place	square Adolphe-et-Jean-Chérioux Père Lachaise Cemetery Père Lachaise Cemetery	square Adolphe-et-Jean-Chérioux Père Lachaise Cemetery PERSON
description	fountain by Brigitte Nahon in Paris, France fountain in Paris, France fountain in Paris, France	fountain by Brigitte Nahon PERSON in Paris GPE, France GPE fountain in Paris GPE, France GPE
material	glass concrete bronze	glass concrete bronze



Fontaine aux roseaux d'argent
fontaine du Fellah
fontaine du Pot-de-Fer



English - en_core_web_sm (v3.5.0)

☒ PERSON	☒ NORP	☒ ORG	☒ GPE	☒ LOC
☒ PRODUCT	☐ EVENT	☐ WORK OF ART	☐ LANGUAGE	
☒ DATE	☐ TIME	☐ PERCENT	☐ MONEY	
☐ QUANTITY	☐ ORDINAL	☐ CARDINAL		

fontaine du Fellah PERSON

fontaine du Pot-de-Fer PERSON



Exercise 4 – Cleaning the Dataset

Cleaning Process

To understand the inconsistencies in the *Genre* column, a text facet was created, revealing many duplicates and variations (e.g., *fountain*, *fountain statue*).

Cleaning involved:

- **Clustering** to merge duplicate or near-duplicate values
- **Standardising formatting** using

After cleaning, the **top three genres** were identified:

1. Sculpture
2. Statue
3. Fountain



Exercise 4 – Cleaning the Dataset (continued)

Analysis

Cleaning removed redundant values and unified formatting, which:

- Makes queries easier and more reliable
- Allows clear identification of top genres
- Enables meaningful before/after comparison
- Supports accurate statistical analysis
- Prepares the *Genre* column for Wikidata reconciliation and Linked Data workflows

Thus, cleaning data is an important step as it improves the workflow of Linked Data (LoD). It also ensures that later reconciliation and querying steps operate on consistent, reliable values, allowing the dataset to integrate more effectively with external knowledge graphs.



Genre Column (Before)

genre change

132 choices Sort by: name count Cluster

- album, work of art, Percent for Art 1
- art museum, sculpture garden 1
- bench, installation 1
- bench, sculpture 1
- bench, sculpture, installation 1
- bench, sculpture, subway
- entrance, installation 1
- bust 1
- cenotaph, sculpture 1
- cenotaph, sculpture, bust 1
- clock, sculpture, automaton 1
- equestrian statue, monumental sculpture 1
- fountain 54

Genre Column (After)

OpenRefine public sculptures csv Pe

Facet / Filter Undo / Redo 3 / 3 <

Refresh Reset all Remove all

genre change

127 choices Sort by: name count Cluster

- sculpture 295
- statue 268
- fountain 54
- sculpture, bust 42
- statue, nude 24
- sculpture, equestrian statue 20
- fountain, sculpture 13
- sculpture, animal sculpture 13
- statue, allegorical sculpture 13
- installation 11
- fountain, drinking fountain 10
- statue, architectural sculpture 10
- mural 9
- war memorial 9
- installation, light art 8
- sculpture, installation 8
- memorial 5
- monument 5
- statue, allegory, allegorical

Exercise 5 – Reconciliation with Wikidata

Matched Creators:

681 creators successfully matched (via *Reconcile* → *Facet* → *By judgment*).

Examples of Matched Entities:

- **Yvonne Serruys** — Q2585677
- **Auguste Biaggi** — Q1835389
- **Max Ingrand** — Q1912755

Creators with 10 Works:

- **Aristide Maillol** — 18 works
- **Anonymous** — 13 works

Analysis

Most creators were matched correctly. However, **70 blank entries** remained unmatched because empty rows were not removed beforehand. Additionally, **235 entries** fell under “None,” requiring manual review. This shows that reconciliation is powerful but not fully automatic — data quality and human oversight remain essential for accurate Linked Data workflows.



Questions	Results
How many creators were successfully matched	681 creators, which was determined by running: <i>Reconcile - Facet - By judgement</i> .
Examples of matched entities (with Wikidata IDs)	Yvonne Serruys (Q2585677) - sculptor (1873-1953) Auguste Biaggi (Q1835389) - French sculptor Max Ingrand (Q1912755) - French stained-glass artist (1908–1969)
Identify creators with more than 10 works	Aristide Maillol 18 Anonymous 13 This was determined by running <i>Column - Facet - Text Facet - Sort by - Count</i> .

Refresh

Reset all

Remove all

☒ creator: judgment [change](#) [invert](#) [reset](#)

3 choices Sort by: name count

(blank) 70

matched 681

[exclude](#)

none 235

Facet by choice counts

Facet / Filter

Undo / Redo 5 / 5



Refresh

Reset all

Remove all

☒ creator

[change](#)

602 choices Sort by: name count

[Cluster](#)

Aristide Maillol 18

anonymous 13

Auguste Rodin 9

Paul Landowski 9

Antoine Bourdelle 8

Emmanuel Frémiet 8

Jean Antoine Injalbert 8

Jules Dalou 7

Charles-François Leboeuf 6

Gustave Michel 6

Hervé Mathieu-Bachelot 6

Jean Beausire 6

Laurent Marqueste 6

Léopold Morice 6

Alexandre Falguière 5



Exercise 6 – Enriching

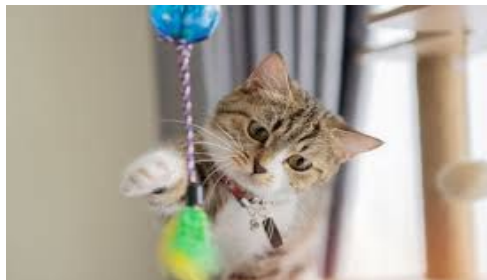


Filtering Approach

Because reconciliation and enrichment were too slow to run on the full dataset, I filtered the Creator column using *Facet* → *Customised facet* → *Word facet*. This gave me a smaller subset of about 200 rows to work with. I then added the Wikidata properties *educated at*, *employer*, and *image* to this subset. Some creators received values, while others did not, indicating that Wikidata coverage varies by person.

Analysis

Even with a small subset, adding information like education, employer, and images makes the dataset more useful. It provides additional context about the creators and makes it easier to check the data. The enrichment also shows how Linked Data can connect different sources and make the dataset easier to analyse, even if some values are missing. It also highlights a common issue: many historical figures lack complete Wikidata entries, so enrichment will always be uneven.



36 total)

Show: 5 10 25 50 100 500 1000 rows

Add columns from reconciled column creator

Add property

Preview

Reset

Suggested properties

award received
child
country of citizenship
educated at
employer
ethnic group
father
field of work
image
member of political party
mother
native language
occupation
place of birth
place of burial

OK

Cancel

Louis Madeline, Léon-Ernest Drivier

fountain, sculpture

square des Anciens-Combattants-d'Indochine,
Palais de la Porte Dorée, porte Dorée

12

Question	Results 
Which Wikidata properties did you add?	<i>educated at (P69), employer (P108), image (P18)</i>
How many rows were successfully enriched?	In the filtered subset of 200 rows, several creators received values for educated at and image , while others remained blank. This shows that enrichment works, but coverage depends on the completeness of each creator's Wikidata entry.
Were there any missing values?	Yes. Many creators lacked values for educated at , employer , or image . This is expected because numerous historical sculptors have incomplete or unevenly curated Wikidata profiles.

Add columns from reconciled column creator

Add property

Suggested properties

award received
child
country of citizenship
educated at
employer
ethnic group
father
field of work
image
member of political party
mother
native language
occupation
place of birth
place of burial

Preview

Reset

creator	award received remove configure	country of citizenship remove configure	educated at remove configure
<not reconciled>			
<not reconciled>			
<not reconciled>			
<not reconciled>			
<not reconciled>			
<not reconciled>			
<not reconciled>			
<not reconciled>			
<not reconciled>			
<not reconciled>			
<not reconciled>			

OK

Cancel

ulpturs csv [Permalink](#)

Extend data at index 3 based on column creator
0% complete [Cancel](#)

5



235 matching rows (986 total)

Remove all

Show as: **rows** records Show: 5 10 25 50 100 500 1000 rows

ange invert reset

exclude

itle	☐ creator	☐ genre	☐ place
ine du Fellah	François-Jean Brallie, Pierre-Nicolas Beauvallet, Jean-François-Théodore Gechter ☒ ☒ Create new item Search for match	statue, fountain	
o of Dantan	Jean-Pierre Dantan, Antoine Laurent Dantan, Édouard Renaud ☒ ☒ Create new item Search for match	tomb, sculptural group	Père Lachaise Cemetery
éric Lemaître	Pierre Granet ☒ ☒ Pierre Granet (100) ☒ ☒ Pierre de Granet (90) ☒ ☒ Jean-Pierre Granet (84) ☒ ☒ Create new item See more Search for match	sculpture, bust	square Frédérick-Lemaître
ine Dejean	Jean-Camille Formigé, Charles Louis Malric ☒ ☒ Create new item Search for match	fountain, drinking fountain	
otteleur	Jacques Perrin ☒ ☒ Create new item Search for match	statue, nude	square Maurice-Gardette

200 matching rows (1,028 total)

Show as: **rows** records Show: 5 10 25 50 100 500 1000 rows

« first < previous 1

▼ All	▼ item	▼ title	▼ creator	▼ educated at	▼ employer	▼ image 2
☆ 	2. http://www.wikidata.org/entity/Q3076272	fontaine du Fellah	François-Jean Bralle, Pierre-Nicolas Beauvallet, Jean-François-Théodore Gechter ☒ ☒ Create new item Search for match			
☆ 	7. http://www.wikidata.org/entity/Q3323374	Pierre Cartellier's tomb	Louis Petitot Choose new match	French Academy in Rome Choose new match		Charles X distribuant des récompenses aux artistes exposants du salon de 1824 au Louvre, le 15 Janvier 1825 (by François Joseph Heim).jpg
☆ 	8. http://www.wikidata.org/entity/Q3323375	Tomb of Dantan	Jean-Pierre Dantan, Antoine Laurent Dantan, Édouard Renaud ☒ ☒ Create new item Search for match			
☆ 	9. http://www.wikidata.org/entity/Q2928779	Frédéric Lemaître	Pierre Granet ☒ ☒ Pierre Granet (100) ☒ ☒ Pierre de Granet (90) ☒ ☒ Jean-Pierre Granet (84) ☒ ☒ Create new item See more Search for match			
☆ 	10. http://www.wikidata.org/entity/Q3076013	fontaine Dejean	Jean-Camille Formigé, Charles Louis Malric ☒ ☒ Create new item Search for match			
☆ 	11. http://www.wikidata.org/entity/Q3076130	fontaine de Charonne	Jean Beausire Choose new match			
☆ 	12. http://www.wikidata.org/entity/Q3076204	fontaine de la Petite-Halle	Jean Beausire Choose new match			
☆ 	21. http://www.wikidata.org/entity/Q3076201	fontaine de la Porte Dorée	Louis Madeline, Léon-Ernest Drivier ☒ ☒ Create new item Search for match			
☆ 	26. http://www.wikidata.org/entity/Q3211751	La Pluie de citations	Jean-Christophe Bailly, Antoine Grumbach ☒ ☒ Create new item Search for match			
☆ 	38. http://www.wikidata.org/entity/Q18145883	Anne de Bretagne	Jean Baptiste Joseph Debay ☒ ☒ Jean Baptiste Joseph Debay (100) ☒ ☒ Jean Baptiste Joseph Debay (100) ☒ ☒ Jean Baptiste Joseph Debay The Elder (84) ☒ ☒ Create new item Search for match			